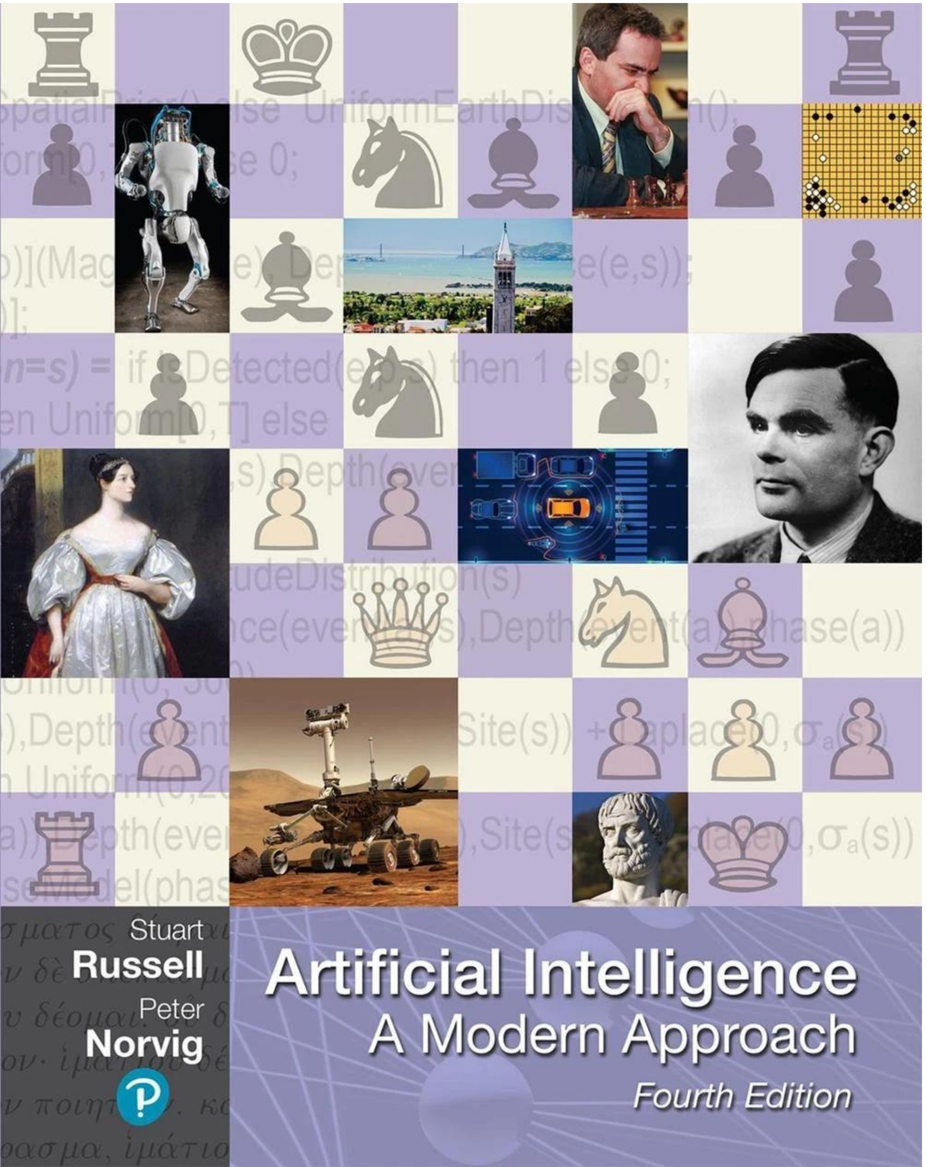
HOA SEN UNIVERSITY

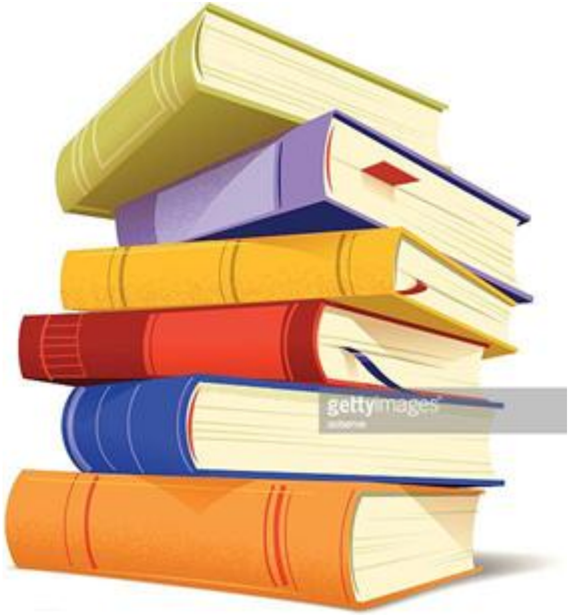
www.hoasen.edu.vn



Chapter 7

An Introduction to Machine Learning

An Introduction to Machine Learning



- Preliminaries
- Applications of Machine Learning algorithms
- The present and the future

HOA SEN UNIVERSITY

www.hoasen.edu.vn

Preliminaries

AI Milestones (1/4)

- 1943: Waren McCulloch and Walter Pitts introduced the first neural network model.
- 1950: Alan Turing posed his famous question: "Can machines think?" and introduced the concept of a test of intelligence.
- 1955: Oliver Selfridge proposed using computers for pattern recognition.
- 1956: The term artificial intelligence was first used at the second Dartmouth conference, organized by John McCarthy, Marvin Minsky, Nathan Rochester of IBM, and Claude Shannon. The term cognitive science also originated in 1956.
- 1957: Frank Rosenblatt invented the first perceptron.

AI Milestones (2/4)

- 1959: John McCarthy invented the LISP programming language.
- 1962: David Hubel and Torsten Wiesel proposed neural networks for computer vision.
- 1964: Joseph Weizenbaum developed Eliza, the first expert system for disease diagnosis.
- 1969: Marvin Minsky and Seymour Papert's book Perceptrons highlighted limitations of neural networks, leading to a funding halt.
- 1979: Paul Werbos proposed the first efficient neural network model with backpropagation, rediscovered in 1986 by Rumelhart, Hinton, and Williams.

AI Milestones (3/4)

- 1987: Terrence Sejnowski and Charles Rosenberg developed NETTalk, an artificial neural network for speech recognition.
- 1987: John H. Holland and Arthur W. Burks invented an adapted computing system capable of learning.
- 1989: Dean Pomerleau proposed ALVINN (autonomous land vehicle in a neural network) for road following.
- 1997: IBM's Deep Blue chess machine defeated world chess champion Garry Kasparov.

AI Milestones (4/4)

- 2011: IBM's Watson defeated Jeopardy! champions Brad Rutter and Ken Jennings.
- 1997-Present: Rapid developments in reinforcement learning, natural language processing, emotional understanding, computer vision, and computer hearing.
- Current Research Trends: Focus on computer vision, hearing, natural language processing, image processing and pattern recognition, cognitive computing, and knowledge representation.

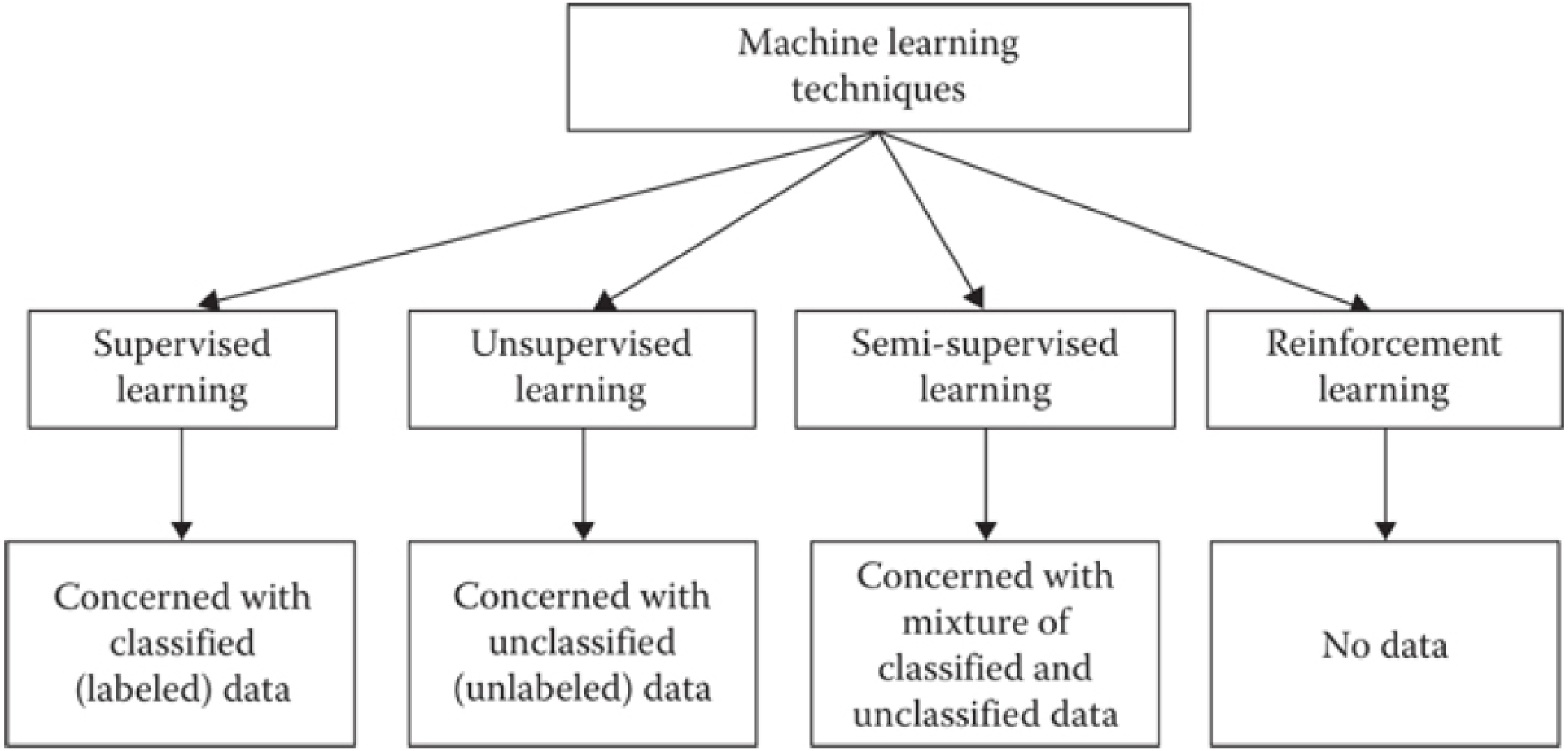
Defining Machine Learning

- The term machine learning means to enable machines to learn without programming them explicitly.
- The main objectives of machine learning are to enable machines to make predictions, perform clustering, extract association rules, or make decisions from a given dataset.

Overview of Machine Learning Techniques (1/2)

- There are four general machine learning methods:
  - Supervised Learning
  - Unsupervised Learning
  - Semi-Supervised Learning
  - Reinforcement Learning

Overview of Machine Learning Techniques (2/2)



HOA SEN UNIVERSITY

The Dataset

Feature vector $\boldsymbol{x}$ (Features, Variables, Attributes)

Class Label y

Examples (Instances, Observation)

Example	Input Attributes										Output
	Alt	*Bar*	*Fri*	*Hun*	*Pat*	*Price*	*Rain*	*Res*	*Type*	*Est*	*WillWait*
$\mathbf{x}_1$	*Yes*	*No*	*No*	*Yes*	*Some*	*\$\$\$*	*No*	*Yes*	*French*	*0–10*	$y_1 = Yes$
$\mathbf{x}_2$	*Yes*	*No*	*No*	*Yes*	*Full*	*\$*	*No*	*No*	*Thai*	*30–60*	$y_2 = No$
$\mathbf{x}_3$	*No*	*Yes*	*No*	*No*	*Some*	*\$*	*No*	*No*	*Burger*	*0–10*	$y_3 = Yes$
$\mathbf{x}_4$	*Yes*	*No*	*Yes*	*Yes*	*Full*	*\$*	*Yes*	*No*	*Thai*	*10–30*	$y_4 = Yes$
$\mathbf{x}_5$	*Yes*	*No*	*Yes*	*No*	*Full*	*\$\$\$*	*No*	*Yes*	*French*	*>60*	$y_5 = No$
$\mathbf{x}_6$	*No*	*Yes*	*No*	*Yes*	*Some*	*\$\$*	*Yes*	*Yes*	*Italian*	*0–10*	$y_6 = Yes$
$\mathbf{x}_7$	*No*	*Yes*	*No*	*No*	*None*	*\$*	*Yes*	*No*	*Burger*	*0–10*	$y_7 = No$
$\mathbf{x}_8$	*No*	*No*	*No*	*Yes*	*Some*	*\$\$*	*Yes*	*Yes*	*Thai*	*0–10*	$y_8 = Yes$
$\mathbf{x}_9$	*No*	*Yes*	*Yes*	*No*	*Full*	*\$*	*Yes*	*No*	*Burger*	*>60*	$y_9 = No$
$\mathbf{x}_{10}$	*Yes*	*Yes*	*Yes*	*Yes*	*Full*	*\$\$\$*	*No*	*Yes*	*Italian*	*10–30*	$y_{10} = No$
$\mathbf{x}_{11}$	*No*	*No*	*No*	*No*	*None*	*\$*	*No*	*No*	*Thai*	*0–10*	$y_{11} = No$
$\mathbf{x}_{12}$	*Yes*	*Yes*	*Yes*	*Yes*	*Full*	*\$*	*No*	*No*	*Burger*	*30–60*	$y_{12} = Yes$

Alternative

Hungry

Patrons

Reservation

Wait time

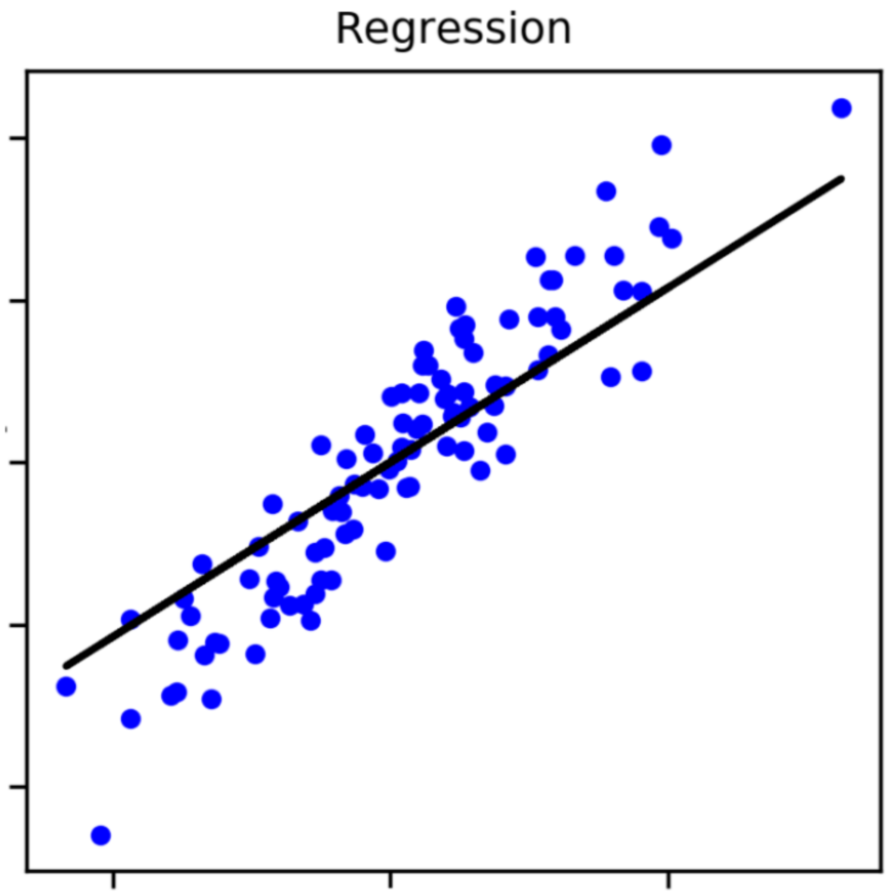
Definition

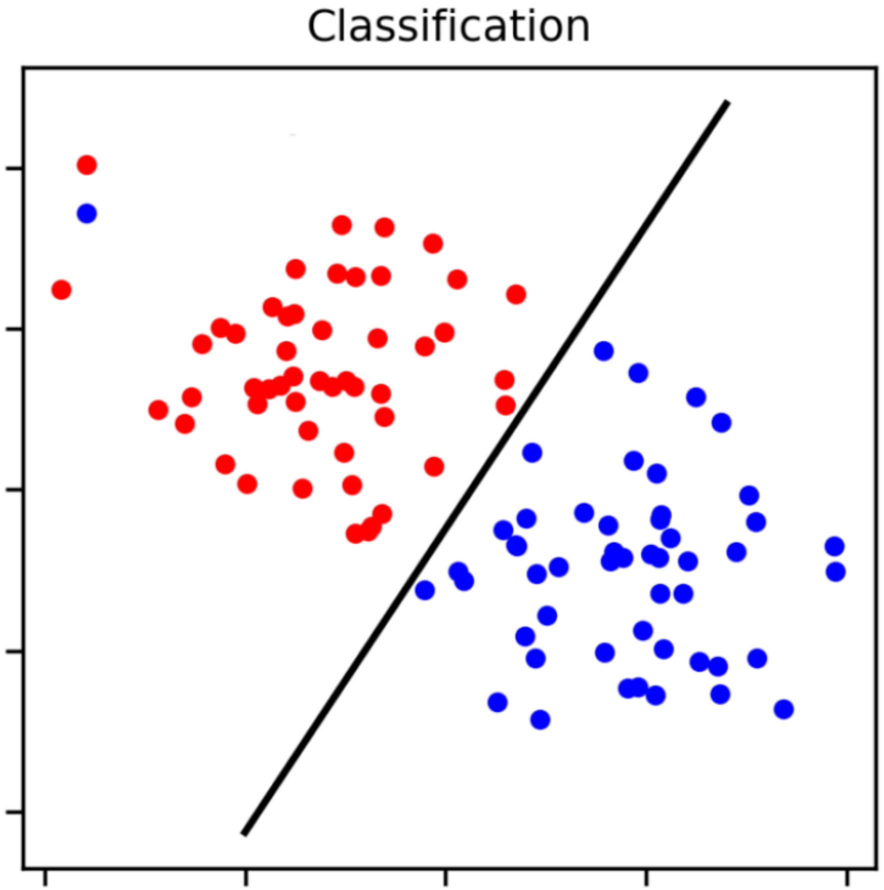
- Definition: In supervised learning, the target is to infer a function or mapping from training data that is labeled.
- Training Data: Consists of an input vector X and an output vector Y of labels or tags.
- A label from vector Y explains its respective input example from input vector X.
- Together, X and Y form a training example.
- If labels do not exist for input vector X, then X is considered unlabeled data.
- Why "Supervised"? The output vector Y's labels are provided by a supervisor, often humans, but sometimes machines.
- Human judgment is generally more expensive but yields higher accuracy compared to machine-labeled data. Manually labeled data is a precious and reliable resource for supervised learning.

Table 1.1 Unlabeled Data Examples along with Labeling Issues

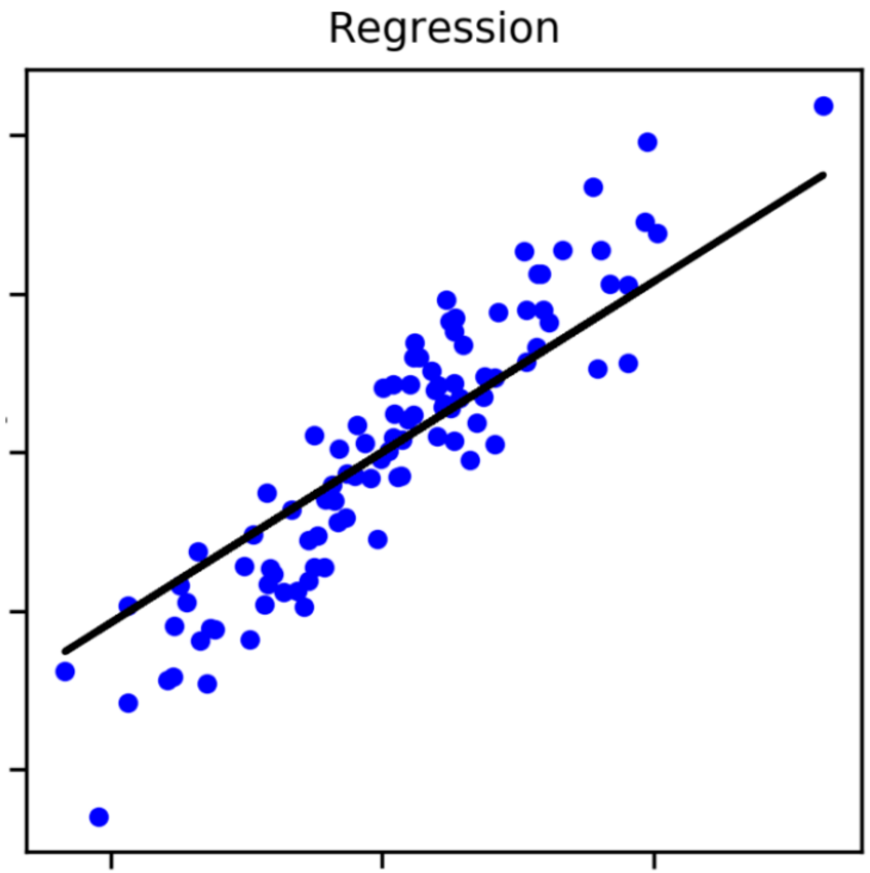
Unlabeled Data Example	*Example Judgment for Labeling*	*Possible Labels*	*Possible Supervisor*
Tweet	Sentiment of the tweet	*Positive/ negative*	Human/ machine
Photo	Contains *house* and *car*	*Yes/No*	Human/ machine
Audio recording	The word *football* is uttered	*Yes/No*	Human/ machine
Video	Are weapons used in the video?	*Violent/ nonviolent*	Human/ machine
X-ray	Tumor presence in X-ray	*Present/ absent*	Experts/ machine

Categories



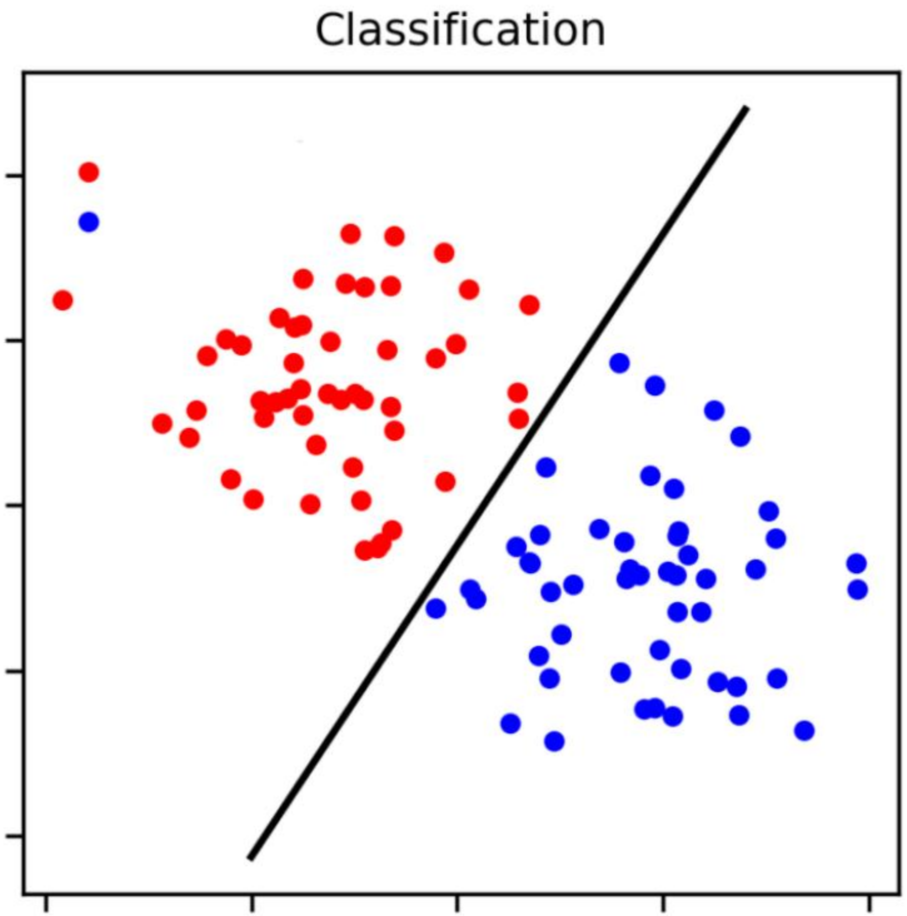


Regression



- The output is a continuous value.
- The model predicts a real number based on input features.
- The objective is to approximate an unknown function that maps inputs to numeric outputs.
- Examples:
  - Predicting house prices based on size and location.
  - Predicting temperature for the next day.
  - Estimating sales revenue for the next month.

Classification



- The output is a discrete class label.
- The model assigns each input example to one class from a finite set of classes.
- The objective is to learn decision boundaries that separate different classes.
- Examples:
  - Classifying emails as spam or not spam.
  - Recognizing handwritten digits from 0 to 9.
  - Medical diagnosis such as disease or no disease.

Definition

- Definition: In unsupervised learning, supervisors or labeled training data are absent.
- Input: All we have is unlabeled data.
- Objective: The core idea is to find a hidden structure within this unlabeled data.
- Reasons for Unlabeled Data:
  - Unavailability of funds for manual labeling.
  - The inherent nature of the data itself.
  - Big Data Context: With numerous data collection devices, data is now collected at an unprecedented rate, characterized by variety, velocity, and volume.
- Extracting insights from this data without a supervisor is a significant challenge for machine learning practitioners today.

Definition

- Definition: This type of learning uses a mixture of classified (labeled) and unclassified (unlabeled) data to generate an appropriate model for data classification.
- Typical Scenario: Labeled data is scarce, while unlabeled data is abundant.
- Objective: To learn a model that can predict classes of future test data better than a model generated using labeled data alone.
- Human Learning Analogy: Our own learning process is similar to semi-supervised learning:
  - Children are supplied with unlabeled data from their environment.
  - They receive labeled data from a supervisor (e.g., a parent teaching names of objects).

Definition

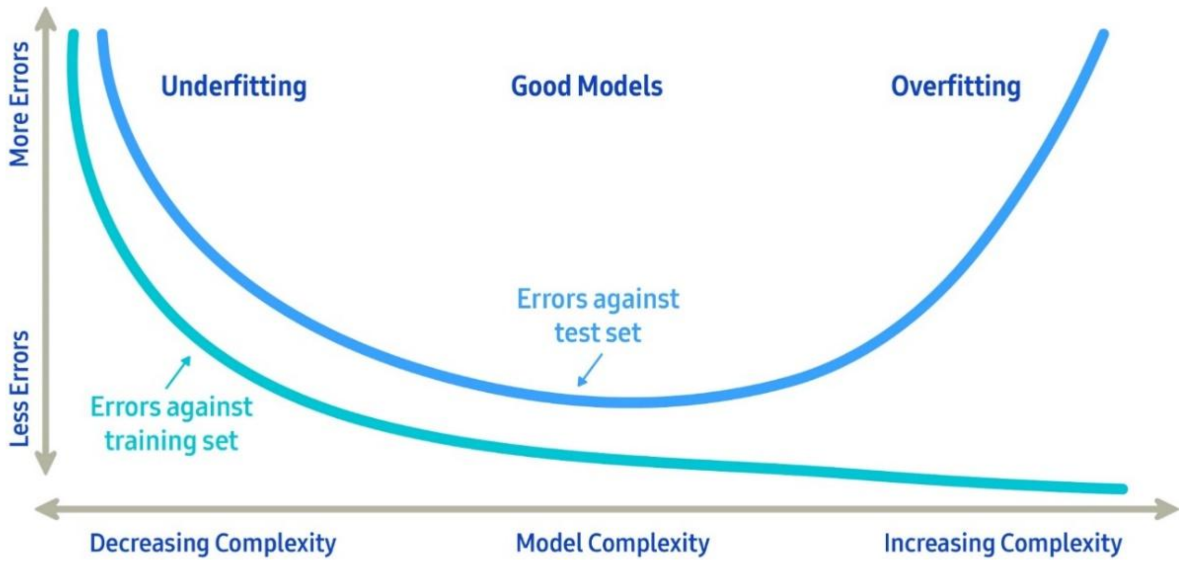
- Definition: The reinforcement learning method aims at using observations gathered from the interaction with the environment to take actions that would maximize the reward or minimize the risk.
- Steps to Produce Intelligent Programs (Agents):
  - Input state is observed by the agent.
  - A decision-making function is used to make the agent perform an action.
  - After the action, the agent receives reward or reinforcement from the environment.
  - State-action pair information about the reward is stored.
  - Policy Fine-Tuning: The stored information helps in fine-tuning the policy for a particular state in terms of action, leading to optimal decision making for the agent.

Validation and Evaluation

- Purpose: Essential for assessing the quality of a model learned from a machine learning algorithm.
- Machine Learning: If a function fits perfectly on training data, the machine learning community questions its performance on testing data.

Overfitting & Underfitting

- Overfitting
  - Overfitting occurs when the model generated tries to fit the training data too closely
  - The model is only good for the training data and will likely produce very poor results with any new test data
  - Basically useless



- Underfitting
  - Underfitting occurs when the model generated has too much deviation from a good predictive model
  - While this model may not be completely useless, the margin of error is too large
- The best machine learning models are simple models that fit the data well

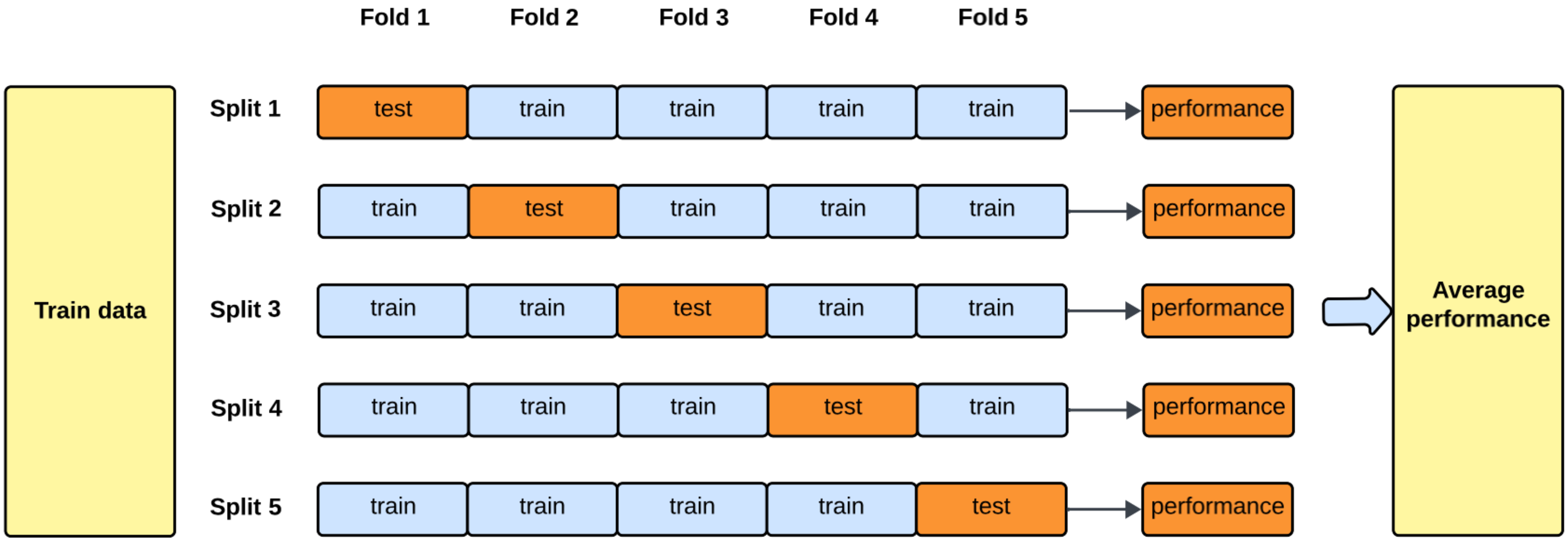
Why Split Data?

- To evaluate model performance fairly
- To prevent overfitting
- To simulate unseen data scenarios
- To tune hyperparameters

How to split data

- The labeled data is splitted into three parts:
  - Training set: used to train/build the model
  - Validation set: used to tune hyperparameters
  - Test set: Used to validate the built model
- Typical Ratios (Train – Validation – Test):
  - 70% – 15% – 15% (Used when data set is medium/large)
  - 60% – 20% – 20% (Used when data set is small)
  - 80% – 20% (Train – Test, with cross-validation and small data set)

Cross-Validation



Cross-Validation

- Cross-validation is especially useful when the available training dataset is quite small and it is not possible to retain a portion of the data just for validation.
- k-Fold Cross-Validation:
  - Split data into k equal folds
  - Repeat k times → each fold used as validation once:
    - Train on **K-1 folds**
    - Validate on the **remaining fold**
  - Compute the **average performance** across all folds
- Ensures more reliable evaluation

HOA SEN UNIVERSITY

www.hoasen.edu.vn

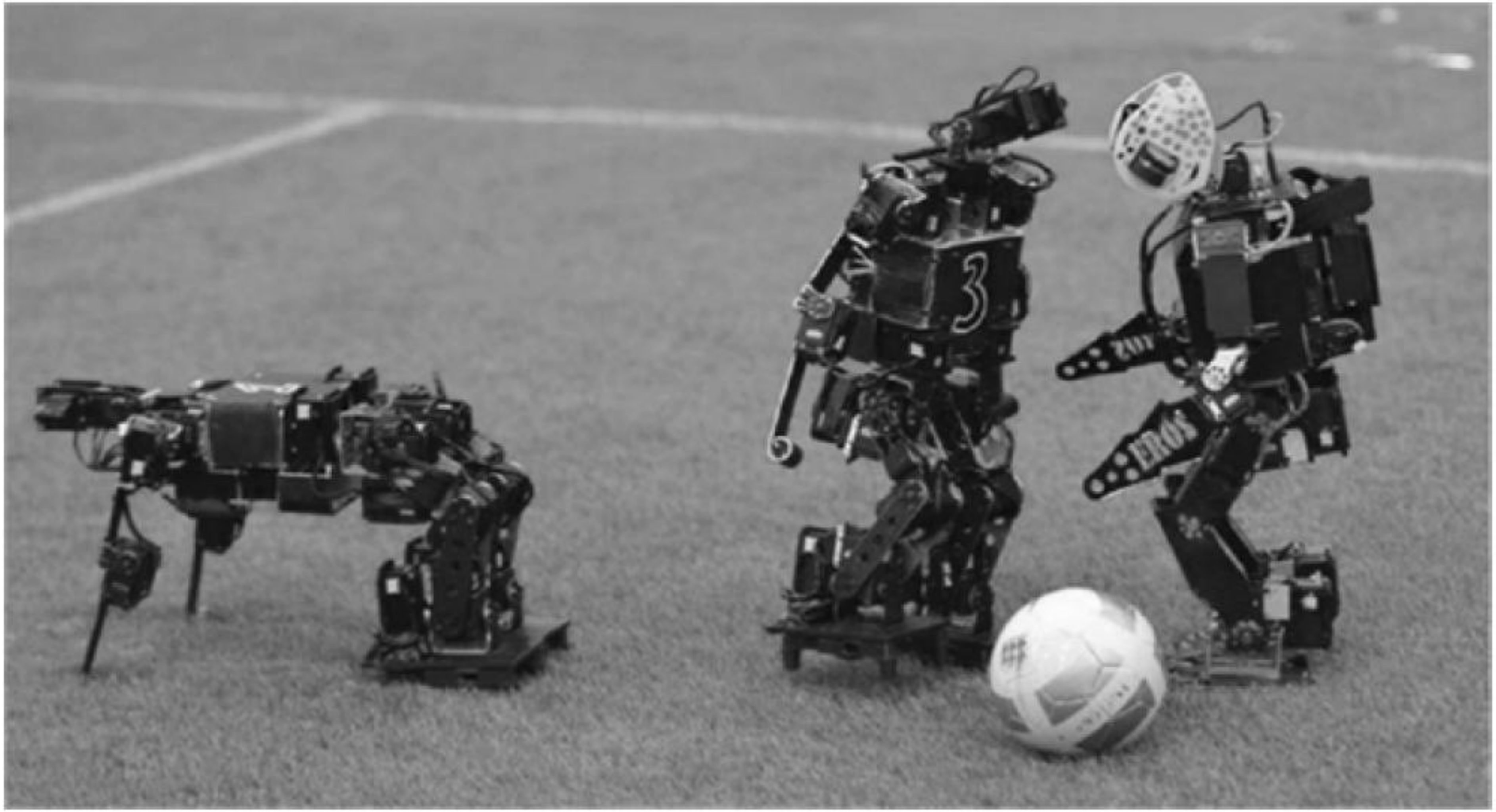
Applications of Machine Learning algorithms

Applications of Machine Learning Algorithms

- Computer vision: image classification, object detection, facial recognition, medical image analysis, and autonomous driving systems.
- Natural language processing: machine translation, chatbots, speech recognition, text summarization, and sentiment analysis.
- Recommendation systems: analyze user behavior and preferences in order to recommend products, movies, music, or news. This is commonly applied in e-commerce platforms and online streaming services.

Applications of Machine Learning Algorithms

- Healthcare: supports disease diagnosis, patient risk prediction, medical image interpretation, drug discovery, and personalized treatment planning based on patient data.
- Finance: fraud detection, credit scoring, risk assessment, and identifying patterns in large-scale financial data.
- Autonomous systems: self-driving cars, robots, and drones. It enables perception, decision-making, and adaptive control in complex environments.
- Business analytics: demand forecasting, customer segmentation, predictive maintenance, and process optimization to improve efficiency and reduce costs.



US and Iranian robot teams competing for RoboCup 2014 final

Toyota tested its self-driving Highway Teammate car on a public road

The Present and the Future

IBM Deep Blue (1/2)

- Historical Achievement: In May 1997, Deep Blue became the first computer system to defeat a reigning chess world champion, Garry Kasparov, in a match.
- Capabilities: Utilized the "brute force of computing power" due to specialized hardware.
- Capable of evaluating 200 million chess positions per second.

IBM Deep Blue (2/2)

- How it Evaluated:
  - Deep Blue's evaluation function was initially written in a generalized form with many customizable parameters.
  - Optimal values for these parameters were determined by the system itself through analyzing thousands of master games.
  - The evaluation function had 8,000 parts, many designed for special positions.
  - Its opening book contained over 4,000 positions and 700,000 grandmaster games.
- Evolution: In 1997, Deep Blue was a dedicated supercomputer. By 2006, chess programs like Deep Fritz (running on a personal computer) could evaluate 8 million positions per second, demonstrating improved software efficiency.

IBM Watson

- Name Origin: Named after IBM's first CEO, Thomas J. Watson.
- Capabilities: A "wonderful machine" capable of answering questions posed in natural language.
- Public Recognition: Gained worldwide fame by winning the first prize in the quiz show "Jeopardy!".
- Impact: With its supercomputing and AI power, Watson helps various industries by powering practical applications.
- Industries Benefiting: Healthcare, finance, legal, and retail sectors.
- Watson is considered "perhaps the most well-known example of artificial intelligence in use today".

Google Now

- Google's innovation, a "landmark for the machine learning world".
- A personal assistant with smartness and intelligence.
- Functions: Answering questions, making recommendations, and performing actions by assigning requests to web services.
- Users can use voice commands for reminders and trivia.
- Proactive Program: Observes user search habits to predict and deliver useful information proactively.

Apple Siri (Speech Interpretation and Recognition Interface)

- A widely used intelligent personal assistant by Apple Inc..
- Supports numerous languages (English, Spanish, French, German, Italian, Japanese, Korean, Mandarin, Russian, Turkish, Arabic).
- Continuously updated to improve responses.
- Context Understanding: Critical for Siri's effectiveness (e.g., reporting a threat instead of showing a map).

Microsoft Cortana

- Another intelligent personal assistant, competing with Google Now and Apple’s Siri.
- Future integrations: Users will soon be able to use Skype to book trips, shop, and plan schedules by chatting with Cortana.

Conclusion

- Machine learning is a rapidly evolving field transforming various aspects of our lives.
- We've covered:
  - The core concept and history of machine learning.
  - Different types of learning: Supervised, Unsupervised, Semi-Supervised, and Reinforcement Learning.
  - Key applications from postal services to healthcare and autonomous vehicles.
  - The current state and future of AI, including famous intelligent systems.
  - The journey towards truly intelligent machines is ongoing, with significant breakthroughs and ethical considerations.


Q
A